

Rohan Kulshrestha

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EDUCATION

Nanyang Technological University

Bachelor of Engineering in Computer Engineering (Expected May 2028)

Singapore

Aug 2023 – Present

Raffles Institution

GCE A Levels

Singapore

Jan 2015 – Dec 2020

TECHNICAL SKILLS

Languages: Python, C++, Java, TypeScript, JavaScript, SQL, C

AI/ML: Machine Learning, RAG, Google Gemini (incl. Vision), Google ADK, LangGraph, OpenAI APIs, Vector Embeddings, NLP, TTS

Web, Mobile and Audio: Next.js, React, React Native, FastAPI, REST APIs, JUCE, VST3/AU plugins, real-time DSP, oversampling

Data and Cloud: PostgreSQL, SQLite, Redis, SQLAlchemy, pgvector, Google Cloud (Cloud Run/SQL/Tasks), Supabase, Docker

Tools: Git, Linux, Pydantic, Asyncio, Playwright, Pandas, NumPy, Figma

EXPERIENCE

Software and AI Engineer Intern

May 2025 – Aug 2025

AlphaGeo

Singapore

- Designed and built an end-to-end ETL pipeline consolidating 11,000+ data center facilities from 6+ sources (Wikidata/SPARQL, OpenStreetMap, PeeringDB, news, reports) with geospatial + fuzzy-match deduplication.
- Developed a LangGraph agent (Gemini 2.5 Pro, Pydantic structured output) that autonomously locates market reports, plus a vision pipeline (Playwright + Gemini Vision) to extract data from interactive, web-only PDF viewers.
- Built a second pipeline tracking the global Tesla Supercharger network with schema-driven field-level change detection (SQLAlchemy reflection) and idempotent upserts for a full audit trail.

PROJECTS

BQST - Audio Mastering Plugin | C++, Real-Time Systems, Concurrency, JUCE | [GitHub](#)

Jun 2026

- Built and shipped a cross-platform native C++ application (VST3/AU/Standalone audio plugin) end to end: DSP core, custom GUI, and a signed, notarized universal macOS installer.
- Implemented the C++ DSP core from scratch (IIR shelving filters and nonlinear waveshaping algorithms for harmonic generation) on a lock-free, allocation-free real-time path with 8x oversampling, meeting the audio thread's hard millisecond deadlines.
- Validated the release with automated testing (pluginval, strictness level 10) across state recall, parameter automation, and sample-rate/buffer-size edge cases.

YourCast - AI Podcast App | Python, Google ADK, Gemini, GCP | [Live Demo](#)

Jun 2025 – Aug 2025

- Built and deployed a serverless full-stack app that generates personalized 3-5 minute AI podcasts, ingesting 16,000+ articles daily from 200+ RSS feeds.
- Implemented semantic deduplication with vector embeddings (cosine-similarity clustering) and multi-agent script generation using Google ADK and Gemini for story selection, structure, and pacing.
- Parallelized text-to-speech across 8 workers for roughly 7x faster audio generation; deployed on Cloud Run, Cloud SQL, and Cloud Tasks, auto-scaling from 0 to 1000+ instances with sub-2-minute generation.

PatentEase - Patent Management App | React Native, TypeScript, FastAPI, Supabase

Mar 2025 – Apr 2025

- Developed a mobile-first patent lifecycle app with an ML-powered similarity checker (vector embeddings and cosine similarity) integrated with Singapore's IPOS database to surface prior art before submission.
- Architected the similarity engine as an independent FastAPI microservice (async REST) for isolated scaling and model swapping; used Supabase for auth, object storage with signed URLs, and real-time updates.

ADDITIONAL EXPERIENCE & ACTIVITIES

Specialist

Jan 2021 – Jan 2023

Singapore Armed Forces

Singapore

- Led and trained a 40-member artillery unit through live-fire operations and mission planning.

Music Production and Audio Engineering

- Composed, produced, mixed, and mastered original songs in Ableton Live; 2M+ streams across platforms.
- Hand-built custom audio gear (a microphone and an optical compressor) from analog circuitry and soldering.